Topology MATM36: Course Summary of Important Results

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Metric Spaces

Definition 1. *Let X be a topological space and $A \subseteq X$. A point $x \in X$ is called an* adherent point *of A if every open neighbourhood U of x satisfies*

$$U \cap A \neq \varnothing.$$

In the special case where X is a metric space with metric d, this is equivalent to requiring that for every $r > 0$,

$$B_r(x) \cap A \neq \varnothing.$$

Definition 2. *Let X be a topological space and $A \subseteq X$. The set A is called* dense *in X if every nonempty open set $U \subseteq X$ satisfies*

$$U \cap A \neq \varnothing.$$

Equivalently, A is dense in X if $\overline{A} = X$.

Definition 3. *Let X be a topological space and $A \subseteq X$. The set A is called* nowhere dense *in X if*

$$\operatorname{int}(\overline{A}) = \varnothing.$$

Baire Category Theorem

Theorem 1 (Baire Category Theorem)**.** *Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^{\infty}$ be a sequence of dense open subsets of X. Then $\bigcap_{n=1}^{\infty} U_n$ is dense in X.*

Proof. Fix a nonempty open ball $B(x, \varepsilon) \subset X$.

Since U_1 is dense and open, choose $y_1 \in U_1 \cap B(x, \varepsilon)$ and $r_1 > 0$ such that

$$\overline{B}(y_1, r_1) \subset U_1 \cap B(x, \varepsilon), \qquad r_1 < 1.$$

Inductively, having chosen y_n, r_n, use that U_{n+1} is dense and open to pick $y_{n+1} \in U_{n+1} \cap B(y_n, r_n)$ and $r_{n+1} > 0$ such that

$$\overline{B}(y_{n+1}, r_{n+1}) \subset U_{n+1} \cap B(y_n, r_n), \qquad r_{n+1} < \frac{1}{n+1}.$$

Then the closed balls are nested:

$$\overline{B}(y_{n+1}, r_{n+1}) \subset \overline{B}(y_n, r_n) \subset \cdots \subset \overline{B}(y_1, r_1) \subset B(x, \varepsilon),$$

and for $m > n$ we have $y_m \in B(y_n, r_n)$, hence $d(y_m, y_n) < r_n \to 0$. Thus (y_n) is Cauchy, so by completeness $y_n \to y \in X$.

For each fixed n, all tails of the sequence lie in $\overline{B}(y_n, r_n)$, so the limit satisfies $y \in \overline{B}(y_n, r_n) \subset U_n$. Hence $y \in \bigcap_{n=1}^{\infty} U_n$, and also $y \in \overline{B}(y_1, r_1) \subset B(x, \varepsilon)$. Therefore

$$B(x, \varepsilon) \cap \bigcap_{n=1}^{\infty} U_n \neq \varnothing,$$

so the intersection is dense. $\square$

Remark 1. *From the definitions it follows that a set $B \subseteq X$ satisfies* $\operatorname{int}(B) = \varnothing \iff X \setminus B$ *is dense in X. In particular, if A is nowhere dense, then $X \setminus \overline{A}$ is open and dense in X.*

Corollary 1. *Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense subsets of X. Then*

$$\operatorname{int}\left(\bigcup_{n=1}^{\infty} F_n\right) = \varnothing.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} \overline{F_n}$ is dense, hence nonempty in every nonempty open set. $\square$

Finite Products of Metric Spaces

Definition 4. *Let (X_k, d_k), $k = 1, \ldots, n$, be metric spaces and set*

$$X = X_1 \times \cdots \times X_n.$$

A metric d on X is said to be a product metric *if convergence in (X, d) is coordinatewise,*

$$x^{(m)} = (x_1^{(m)}, \ldots, x_n^{(m)}) \to x = (x_1, \ldots, x_n)$$

in X if and only if for each $k = 1, \ldots, n$,

$$x_k^{(m)} \to x_k \quad \text{in } X_k.$$

Theorem 2. *Suppose d is a metric on $X = X_1 \times \cdots \times X_n$ satisfying the above coordinatewise convergence property. Then the open sets in (X, d) are precisely the unions of product sets*

$$U_1 \times \cdots \times U_n,$$

where each U_k is open in X_k.

Proof. First, if $E_k \subset X_k$ is closed for each k, then

$$E_1 \times \cdots \times E_n$$

is closed in X. Indeed, if $x^{(m)} \in E_1 \times \cdots \times E_n$ and $x^{(m)} \to x$ in X, then by coordinatewise convergence, $x_k^{(m)} \to x_k$ in X_k. Since each E_k is closed, $x_k \in E_k$, hence $x \in E_1 \times \cdots \times E_n$.

Now let $U_k \subset X_k$ be open. Then

$$V^{(k)} := X_1 \times \cdots \times X_{k-1} \times U_k \times X_{k+1} \times \cdots \times X_n$$

is open in X, since its complement is of the form

$$X_1 \times \cdots \times (X_k \setminus U_k) \times \cdots \times X_n,$$

which is closed by the previous remark. Hence

$$U_1 \times \cdots \times U_n = \bigcap_{k=1}^{n} V^{(k)}$$

is open in X, and any union of such products is open.

Conversely, let $U \subset X$ be open and $x = (x_1, \ldots, x_n) \in U$. We show there exist open sets $U_k \subset X_k$ such that

$$x \in U_1 \times \cdots \times U_n \subset U.$$

Suppose not. Then for each $m \in \mathbb{N}$,

$$B_{X_1}(x_1, 1/m) \times \cdots \times B_{X_n}(x_n, 1/m) \not\subset U.$$

Thus there exists $x^{(m)} \notin U$ with

$$d_k(x_k^{(m)}, x_k) < \frac{1}{m}, \qquad k = 1, \ldots, n.$$

Then $x_k^{(m)} \to x_k$ for each k, hence $x^{(m)} \to x$ in X. Since $X \setminus U$ is closed and each $x^{(m)} \in X \setminus U$, we conclude $x \in X \setminus U$, a contradiction. Therefore such open sets U_k exist, and U is a union of product sets. $\square$